SYSDESIGN REFERENCE

Design a URL Shortener

Requirements-to-architecture reference walkthrough.

Map short, globally unique keys to destination URLs under read-heavy traffic.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / URL-SHORTENER

SYSDESIGN

Design a URL Shortener

CORE IDEA

Map short, globally unique keys to destination URLs under read-heavy traffic.

Reference architecture decision chain

01	02	03	04
Id Generation	Encoding	Redirect Semantics	Caching

Decisions to defend

+ ID generation
+ encoding
+ redirect semantics
+ caching
+ abuse controls
+ and analytics.

Failure modes to test

+ Hot links
+ malicious destinations
+ key collisions
+ and expired redirects.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a URL Shortener

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

REVIEW GATE: TECHNICAL REVIEW. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.